

Deep Learning-Based ECG Arrhythmia Detection System: A Comparative Study of CNN, RNN, and LSTM Architectures

1st Patiyuth Pramkeaw
Media Technology Program
King Mongkut's University of
Technology Thonburi
Bangkok 10140, Thailand
patiyuth.pra@kmutt.ac.th

2nd Warit Leangruttananon
Media Technology Program
King Mongkut's University of
Technology Thonburi
Bangkok 10140, Thailand
zernmcp@gmail.com

3rd Pimyada Suksomboon
Media Technology Program
King Mongkut's University of
Technology Thonburi
Bangkok 10140, Thailand
pr.pim.ray@gmail.com

4th Tawan Tapianthong
Media Technology Program
King Mongkut's University of
Technology Thonburi
Bangkok 10140, Thailand
tawan.tapi@mail.kmutt.ac.th

5th Setthawut Kitdanarakorn
Media Technology Program
King Mongkut's University of
Technology Thonburi
Bangkok 10140, Thailand
setthawut36755@gmail.com

Abstract—CVDs remain the leading cause of death in the world with an estimated 17.9 million deaths annually. Although the electrocardiogram (ECG) analysis is the most basic element of arrhythmias detection, manual analysis is a time-consuming and inaccurate process. To overcome this, we came up with an effective detection architecture that can be used to analyze the effectiveness of four different types of deep learning structures, namely the Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent neural network (RNN) and the Long short term memory (LSTM) networks. Our research uses the MIT-BIH Arrhythmia Database to establish a large performance difference between the conventional sequential models and hybrid ones. We find that a single CNN-LSTM model performs better than their individual counterparts with a classification accuracy of more than 98. The hybrid method is effective in capturing both spatial morphological and temporal dependencies in the ECG signal to be able to identify normal, supraventricular, ventricular and fusion beats with a high degree of accuracy. In addition to raw accuracy, the model has computational efficiency that can be implemented in a clinical environment in real-time. This study highlights how deep learning can help decentralize expert cardiac diagnostics, which can offer a scalable solution to healthcare professionals operating in resource-constrained settings.

Keywords—Arrhythmia detection, Electrocardiogram (ECG), Convolutional Neural Networks, Cardiovascular disease diagnosis

I. INTRODUCTION

The health issue of cardiovascular diseases (CVDs) is the greatest worldwide, with the death rate being 17.9 million per year and 31% of the total worldwide mortality [1]. Mortality rates have been increasing alarmingly in Thailand due to CVD, especially in the mortality rates of ischemic heart disease; the rate of mortality was increasing by 22.4 to 27.8 per 100,000 population, and the cerebrovascular disease mortality rate was by 33.4 to 40.9 per 100,000 population [2]. The financial impact is huge and the cost of treating cardiac patients in Thailand is more than 6.9 billion baht annually, which affects individuals and their families as well as the healthcare system in the country. Cardiac arrhythmias, which are abnormal heart rhythms, are a life-threatening group of CVDs to be detected and treated. These disorders may appear in the form of several malfunctions such as valve problems in the heart, stroke, and heart attack, and the symptoms may vary

such as the chest pains and fatigue to the sudden cardiac arrest [3]. Predictive data mining techniques have also been applied to stroke disease, demonstrating the potential of computational methods for cardiovascular risk assessment [37]. Nevertheless, a significant number of patients have no symptoms, and it is difficult to detect them, but it is important to prevent unfavorable consequences. Electrocardiogram (ECG) is still the gold standard of non-invasive diagnostic tools of cardiac arrhythmia detection and characterization. The interpretation of the traditional ECG depends largely on the visual interpretation of the expert cardiologists, is time consuming, prone to inter-observer inaccuracies and does not suit real time monitoring uniform system over a long period. Holter monitoring which captures continuous ECG signals in the 24-48 hours form gives rise to large volumes of data that cannot be effectively analyzed without automated analysis tools to detect clinically significant arrhythmic events [4]. In the recent past, deep learning has transformed medical signal processing and presented new opportunities to automated ECG analysis. Beyond cardiac applications, machine learning frameworks have been successfully applied to the detection of other chronic diseases such as diabetes [35]. DNNs, specifically Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have been shown to exhibit impressive performance as far as learning hierarchical feature representations without feature engineering (only with raw or slightly preprocessed ECG data) is concerned [5], [6], [7]. The accuracy rates of these methods have been demonstrated to outperform most benchmark datasets with above 98% accuracies, and even surpass human expert performance on some classification tasks [8], [9], [10]. Nevertheless, even with such a great progress, a number of difficulties are still present in the creation of clinically deployable ECG arrhythmia detection systems. These involve dealing with the problem of class imbalance in arrhythmia data, the provision of generalization to other patient groups (inter-patient variability), the provision of real-time processing features to continuous monitoring applications and interpretability to be accepted by clinicians [11], [12], [13]. Moreover, the available research is primarily dedicated to the assessment of individual architecture, which cannot be used to conduct extensive comparative analyses to inform the selection of the most effective architecture to apply to a certain clinical case. In this paper, these gaps are taken

care of by detailing a deep learning-based ECG arrhythmia detection system that systematically analyzes and compares various neural network architectures such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN) and Long Short-Memory (LSTM) networks. We contribute to it namely: Extensive Architecture Comparison: We give a per-architectural analysis of four big architectures of deep learning (ANN, CNN, RNN, LSTM) and their combinations on the same dataset and experimental regime, which allows one to compare their performance directly. Hybrid Model Development: We design and test hybrid CNN-LSTM models that exploit the spatial feature-extraction potential of CNN with the time-dependency modeling of LSTM, and outperform single-architecture models. Clinical Validation: we are tested on the MIT-BIH Arrhythmia Database, the best-known benchmark in the research on cardiac arrhythmia and ensure reproducibility and comparability with the existing literature. Guidelines to implementation: Our implementation has detailed specifications of implementation, such as preprocessing pipelines, network architecture and training protocol that eases replication and clinical translation. The rest of the paper will take the following structure: Section 2 will conduct a review of the related work in the field of ECG arrhythmia detection with the help of deep learning. Section 3 explains our methodology, such as the description of the datasets, preprocessing procedures, and neural network structures. In Section 4, the experimental results and the analysis of comparative performance are provided. Section 5 is a discussion of findings, clinical implications and limitations. Futuristic researches are closed with section 6.

II. RELATED WORK

2. Work that is related

In recent years, the use of deep learning to find arrhythmias in ECGs has grown rapidly. Many studies have shown that neural network-based methods are better than traditional machine learning methods. This part talks about recent improvements in deep learning architectures for ECG analysis, grouped by architectural style.

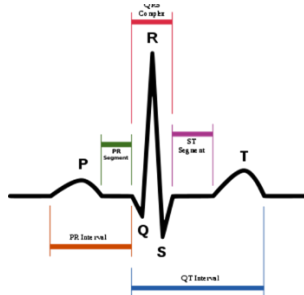


Fig. 1. Normal electrocardiogram

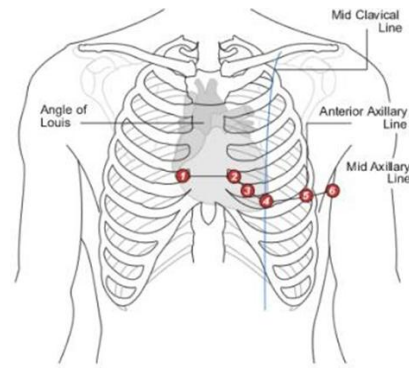


Fig. 2. Comparison of ECG Classification Methods

2.1 Methods for Convolutional Neural Networks

Convolutional Neural Networks have become the leading architecture for ECG signal analysis because they can automatically learn hierarchical spatial features from raw or only slightly processed signals. Odugoudar et al. [8] created a six-layer CNN with residual blocks that could tell five different types of arrhythmia (LBBB, RBBB, APC, PVC, and Normal Beat) on the MIT-BIH dataset with 98.2% accuracy. The residual block architecture solves the problem of the vanishing gradient in deep networks, which makes training more effective.

Azeroual et al. [14] showed that CNN-based classification could correctly identify 99.24% of the time, 99.24% of the time, and 96.94% of the time for four types of arrhythmia (N, SVEB, VEB, F) in the MIT-BIH database. Their work focused on how automated classification could cut down on the time it takes to make a diagnosis. Hu [15] also suggested a 10-layer one-dimensional CNN that was 1-3% better than previous methods on all metrics, with 99.43% accuracy, 97.86% sensitivity, and 99.64% specificity.

Akbar et al. [16] solved the important problem of class imbalance by adding data from the MIT-BIH supraventricular arrhythmia database to the MIT-BIH arrhythmia database. Their 1D-CNN with deep residuals got 99.10% accuracy for 16 multiclass categories and 99.25% accuracy for 5 subject-class categories. This is a lot better than what other studies have found. Li et al. [17] integrated the Taguchi method with CNNs for parameter optimization, illustrating that methodical hyperparameter tuning substantially influences classification performance. Furthermore, CNNs have demonstrated effectiveness in other medical detection domains, such as the systematic development of LeNet-based models for cataract detection [34], confirming the versatility of convolutional architectures for clinical diagnosis tasks. Similarly, machine learning approaches including bootstrap aggregating have been applied to bio-signal classification problems such as electrical impedance-based breast tissue classification [36], highlighting the broader applicability of computational methods to biomedical signal analysis.

2.2 LSTM and Recurrent Network Methods

Long Short-Term Memory networks are great at finding temporal dependencies in sequential data. This makes them perfect for analyzing ECG signals, where temporal patterns are important for clinical purposes. Malleswari et al. [18] created a DWT-BiLSTM classifier that correctly classified five different types of heartbeats 98% of the time, 96.9% of

the time, and 97.5% of the time. Their method used Discrete Wavelet Transform (DWT) to clean up ECG signals before classifying them.

Degachi et al. [19] put forth an attention-augmented bidirectional LSTM-based encoder-decoder architecture that attained 99.87% accuracy on the MIT-BIH database. They came up with new ideas like using a Channel Attention Module (CAM) to get rid of unnecessary features and Adaptive Synthetic Sampling (ADASYN) to fix class imbalance. The attention mechanism made it easier for the encoder-decoder to focus on clinically important signal segments.

Kuila et al. [20] merged the Differential Evolution Algorithm (DEA) with LSTM for feature optimization and arrhythmia detection, whereas Singh et al. [21] combined Recurrent CNN with Grey Wolf Optimization (GWO) to improve classification performance. These studies show that more and more people are using metaheuristic optimization algorithms with deep learning to make models work better.

2.3 Hybrid CNN-LSTM Structures

Many recent studies have looked into hybrid architectures that combine spatial feature extraction with temporal dependency modeling. This is because CNNs and LSTMs have different strengths that work well together. Swaroop et al. [22] created a hybrid CNN-LSTM model that got 98.94% accuracy on the MIT-BIH dataset. This model worked better than CNN-only or LSTM-only models. Their design used CNNs to find spatial patterns in ECG segments and LSTMs to find relationships between heartbeats over time.

Yadav et al. [23] suggested an improved ResNet-50 model that included Conv-1D and LSTM layers. It got 98.7% accuracy and 0.06 Mean Squared Error (MSE) on the MIT-BIH database. The ResNet-50 backbone made it possible to extract deep features while keeping the gradient flow going through residual connections. Hassan et al. [24] performed a systematic optimization study of CNN-LSTM architectures, examining the effects of different filters, kernel sizes, and layer depths. The Optimized CNN (OCNN) was 99.0% accurate, and the Optimized CNN-LSTM (OCNN-LSTM) was 96.1% accurate. This gave us useful information about the trade-offs in architecture design.

Eleyan et al. [25] created a deep learning framework that uses CNNs and LSTMs to classify ECG signals from both the MIT-BIH and BIDMC databases in two different ways (five-class and three-class classification). This shows that the architecture can work with different types of classification. Mrudhhula et al. [26] stressed how important advanced signal processing is in hybrid models. They showed that the quality of preprocessing has a big effect on how accurate the classification is, especially for rare types of arrhythmia.

2.4 New Architectures and Advanced Attention Mechanisms

More and more recent research has used attention mechanisms to make models easier to understand and better at what they do. Li et al. [27] put forward a multi-branch CNN that uses different receptive fields along with a Channel Attention Module (CAM) and a Bidirectional LSTM (BLSTM). This combination achieved a sensitivity of 98.5% and an F1-score of 98.2% for Ventricular Ectopic Beats (VEB). The multi-branch architecture made it possible to get features at different scales, and the attention mechanism got rid of extra information.

Najia et al. [28] created a better hybrid model that combined CNN, BiLSTM, and Convolutional Block Attention Module (CBAM) for classifying ECG segments. The model had an accuracy of 99.20%, a sensitivity of 97.50%, a specificity of 99.81%, and a mean F1-score of 98.29%. Their framework used the Synthetic Minority Over-sampling Technique (SMOTE) to fix the class imbalance, and it worked well for all types of arrhythmias.

Liu [29] presented ECGBERT, an innovative hybrid model that integrates 1D-CNN with BERT (Bidirectional Encoder Representations from Transformers) for advanced ECG arrhythmia classification. This new method uses CNN's ability to extract local features and BERT's ability to model global context through Transformer mechanisms. It can find long-range dependencies between abnormal heartbeats without needing hand-crafted features.

Daydulo et al. [30] optimized pretrained ResNet-50 and AlexNet models for the classification of ECG signals into cardiac arrhythmia, congestive heart failure, and normal sinus rhythm, attaining an accuracy of 99.2%. Their transfer learning method showed that computer vision models that have already been trained can be used for medical signal processing tasks.

2.5 Strategies for Ensemble and Hybrid Learning

Ensemble learning methods have become more popular because they make classification more accurate and reliable. Vijayalakshmi et al. [31] investigated ensemble methodologies that amalgamate CNN, LSTM, BiLSTM, GRU, SVM, LightGBM, and Random Forest models, with a Voting Classifier that integrates various boosted models, resulting in substantial enhancements in classification accuracy. Ahuja et al. [32] created a hybrid CNN-Random Forest (CNN-RF) model that got 95.2% accuracy, 94.6% precision, 94.1% recall, and 94.3% F1-score. This shows that combining deep learning with traditional machine learning classifiers can make them easier to understand while still getting good results.

Zeng et al. [33] integrated Tunable Q-factor Wavelet Transform (TQWT) and Complete Ensemble Empirical Mode Decomposition (CEEMD) with deep CNN-LSTM networks for arrhythmia detection, underscoring the significance of sophisticated signal decomposition methods in preprocessing workflows.

2.6 Research Gaps and Prospects

Even though a lot of progress has been made, there are still some research gaps. First, the majority of studies concentrate on single-database evaluation, primarily MIT-BIH, which constrains the assessment of generalizability across varied patient populations and recording conditions. Second, class imbalance is still a big problem. Many studies show that the overall accuracy is high, but the performance on rare arrhythmia types is low. Third, there isn't much focus on real-time implementation issues, like the need for low latency and high computational complexity in applications that need to be monitored all the time. Fourth, deep learning models' lack of interpretability and explainability is still a major barrier to clinical use, and there aren't many studies that give clinically useful explanations for model predictions.

We fill in these gaps by doing a full comparative evaluation of several architectures under the same experimental conditions, using advanced preprocessing and class balancing techniques,

and giving detailed implementation specifications for clinical translation. In the next section, we go into great detail about how we did our work.

III. METHODOLOGY

This section details the complete process we used to create and test our deep learning-based ECG arrhythmia detection system. We go into detail about the dataset's features, the preprocessing pipeline, the neural network architectures, and the experimental protocols.

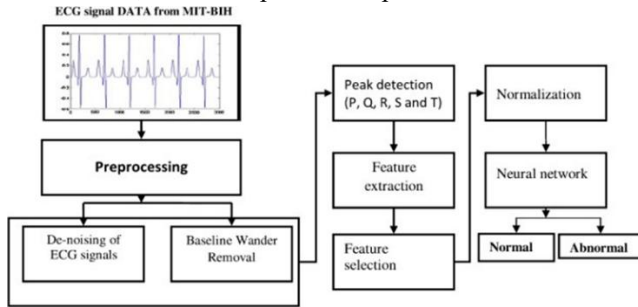


Fig. 3. MIT-BIH Database Overview

3.1 Dataset Description

We used the MIT-BIH Arrhythmia Database, which is the most well-known benchmark dataset for research on cardiac arrhythmia. This made sure that our results could be repeated and compared to other studies. The MIT-BIH database has 48 half-hour clips of two-channel ambulatory ECG recordings from 47 people who were studied by the BIH Arrhythmia Laboratory between 1975 and 1979. The recordings were digitized at a rate of 360 samples per second per channel, with an 11-bit resolution and a range of 10 mV.

Features of the dataset:

There are a total of 48 recordings, numbered 100-124 and 200-234.

- Length: Each recording lasts 30 minutes.
- Rate of sampling: 360 Hz
- Resolution: 11 bits (0.5 mV for each bit)
- Leads: Two channels, usually Modified Lead II and V1, V2, V4, or V5
- Annotations: Expert cardiologists wrote notes on each beat.
- Total Beats: About 110,000 heartbeats with notes

The database has a wide range of arrhythmia types that are based on real-life clinical situations. We put the 15 original beat types into five main groups based on the Association for the Advancement of Medical Instrumentation (AAMI) EC57 standard:

1. Normal (N): Normal beats and beats with a left/right bundle branch block
2. Supraventricular Ectopic Beat (SVEB/S): Atrial premature beats, aberrated atrial premature beats, nodal premature beats, and supraventricular premature beats.
3. Ventricular Ectopic Beat (VEB/V): early contraction of the ventricles, escape beats in the ventricles

4. Fusion Beat (F): The joining of normal beats and ventricular beats
5. Unknown Beat (Q): beats that are paced, a mix of paced and normal beats, or beats that can't be put into any of the other categories

Class Distribution: The dataset shows a big class imbalance, which is a common problem with medical datasets:

- Normal (N): about 75,000 beats (68%)
- SVEB (S): about 2,800 beats, or 2.5%
- VEB (V): about 7,100 beats (6.4%)
- Fusion (F): about 800 beats (0.7%)
- Unknown (Q): about 8,000 beats (7.3%)

3.2 Preprocessing of Signals

For deep learning models to work well, preprocessing is very important because ECG signals can be affected by many different types of noise, such as baseline wander, powerline interference, muscle artifacts, and electrode motion artifacts. The following steps make up our preprocessing pipeline:

3.2.1 Filtering Out Noise

We used a multi-stage filtering method to get rid of different types of noise:

1. Baseline Wander Removal: A high-pass Butterworth filter with a cutoff frequency of 0.5 Hz gets rid of low-frequency baseline drift that happens when you breathe and move your body.
2. Removing Powerline Interference: A notch filter at 50/60 Hz gets rid of powerline interference, which is a common problem with ECG recordings.
3. To get rid of high-frequency muscle artifacts and electrode noise while keeping clinically important ECG parts, use a low-pass Butterworth filter with a cutoff frequency of 40 Hz.

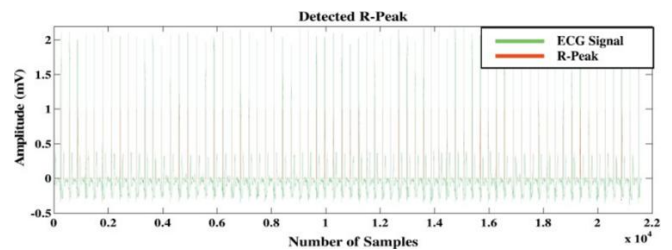


Fig. 4. ECG Signal Preprocessing Pipeline

Fig.4. shows that finding the R-peak is the most important part of working with an electrocardiogram (ECG) (Manikandan and Soman 2012). The R-peak is a clear waveform, so it is usually the first thing to be checked. The first step is to choose the right wavelet coefficients. Using the discrete wavelet transform method, you can find O, S, T, and P once you find the R-peak and adjust the coefficients correctly.

3.2.2 Normalization of Signals

We used Z-score normalization on each ECG segment to make sure the mean was zero and the variance was one:

$$x_{\text{normalized}} = (x - \mu) / \sigma \quad (1)$$

3.2.3 Finding and dividing the R-Peak

For heartbeat segmentation, it is very important to be able to accurately find the R-peak. We used the Pan-Tompkins algorithm, which is a common way to find QRS in real time, and we did the following:

1. Bandpass Filtering: A 5–15 Hz bandpass filter to make the QRS complex stronger
2. Differentiation: Focus on the high-frequency parts of QRS
3. Squaring: Make the QRS complex louder compared to the noise
4. Moving Window Integration: Make the signal smoother
5. Adaptive Thresholding: Find R-peaks using thresholds that change over time

After finding the R-peaks, we took fixed-length segments that were centered on each R-peak. We took 180-sample windows (0.5 seconds) centered on each R-peak based on the 360 Hz sampling rate and the average QRS duration. This gave us the whole PQRS complex with enough context.

3.2.4 Adding More Data

We used a number of data augmentation methods to fix the class imbalance:

1. Fake Minority The Over-sampling Technique (SMOTE) makes fake samples for minority classes by interpolating between existing samples in feature space.
2. Time Warping: Randomly stretches or compresses time in ways that are physiologically possible ($\pm 5\%$).
3. Amplitude Scaling: Changing the amplitude randomly by up to 10% to mimic changes in recording gain.
4. Adding Gaussian Noise: This makes the model more stable by adding small amounts of Gaussian noise (SNR > 20 dB).

3.3 Architectures for Deep Learning

We carefully looked at four major neural network architectures and the different ways they could be combined. This part goes into great detail about each architecture.

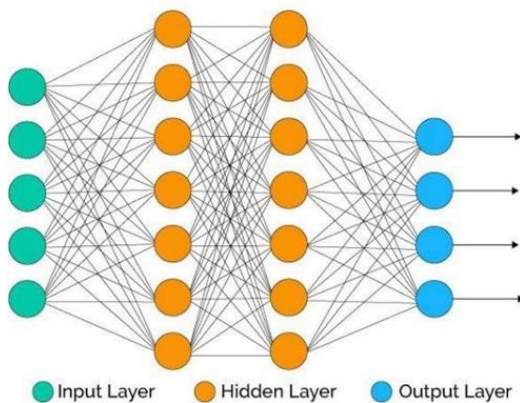


Fig. 5. Components of an artificial neural network

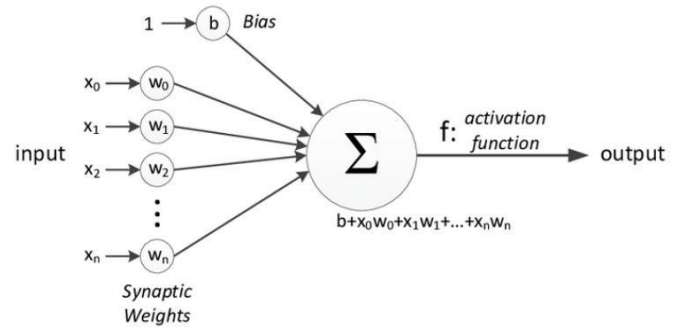


Fig.6. shows how an artificial neural network works. It consists of input data that comes in through synapses. These synapses have different weights that show how strong each node is and biases that help the activation function work better.

$$y = f \left[\left(b + \sum_{i=1}^n (x_i w_i) \right) - T \right] \quad (2)$$

3.3.1 Artificial Neural Network (ANN)

We used a fully connected Artificial Neural Network as a baseline model. The structure is made up of:

- Input Layer: 180 neurons (one for each of the 180-sample ECG segments)
- ReLU activation for 128 neurons in Hidden Layer 1
- ReLU activation for 64 neurons in Hidden Layer 2
- 32 neurons with ReLU activation make up Hidden Layer 3.
- Output Layer: 5 neurons with Softmax activation (for classifying into 5 groups)
- Dropout: To keep the model from overfitting, there is a 0.5 dropout rate after each hidden layer.
- Total Number of Parameters: ~30,000

The ANN is a starting point for showing how more advanced architectures that can capture spatial and temporal patterns are better.

3.3.2 Convolutional Neural Network (CNN)

Our CNN architecture uses convolutional layers to automatically learn hierarchical spatial features from ECG signals.

Architecture Specification:

- Input: 180×1 (one-lead ECG segment)
- Conv1D Layer 1: 64 filters, kernel size 5, stride 1, and ReLU activation
- Layer 1 of MaxPooling1D has a pool size of 2.
- Conv1D Layer 2 has 128 filters, a kernel size of 5, a stride of 1, and ReLU activation.
- MaxPooling1D Layer 2: Size of the pool is 2
- Conv1D Layer 3 has 256 filters, a kernel size of 3, a stride of 1, and ReLU activation.
- MaxPooling1D Layer 3: The pool size is 2.
- Layer that flattens
- Dense Layer 1: 128 neurons with ReLU activation
- 0.5% of people drop out
- Softmax activation for Dense Layer 2 (Output): 5 neurons
- Total number of parameters: ~250,000

The convolutional layers gradually pull out higher-level features. The first layers learn basic waveform patterns like P, QRS, and T waves, while the deeper layers learn more complex patterns that are specific to arrhythmias.

3.3.3 Recurrent Neural Network (RNN)

We used a standard RNN architecture to find time-based connections in ECG sequences:

Architecture Specification:

- Input: 180×1 (sequential ECG samples)
- RNN Layer 1: 128 units, sequences that come back
- 0.3% dropout
- Layer 2 of the RNN has 64 units.
- 0.3% of the time
- 32 neurons in Dense Layer 1, and ReLU activation
- Dense Layer 2 (Output): 5 neurons with Softmax activation
- There are about 85,000 parameters in total.

RNNs can theoretically capture long-term dependencies, but in practice, they have problems with vanishing gradients, which makes them less useful for long sequences.

3.3.4 Long Short-Term Memory (LSTM)

LSTM networks use gating mechanisms to fix the vanishing gradient problem that regular RNNs have.

Specification for Architecture:

- Input: 180×1 (ECG samples taken one after the other)
- LSTM Layer 1 has 128 units and returns sequences.
- Dropout: 0.3
- LSTM Layer 2 has 64 units.
- 0.3% of people dropped out
- 32 neurons in Dense Layer 1, and ReLU activation
- Softmax activation for Dense Layer 2 (Output): 5 neurons
- Total number of parameters: $\sim 120,000$

The LSTM's forget gate, input gate, and output gate let it choose what information to keep and what to forget. This makes it better at finding long-term patterns in ECG signals over time.

3.3.5 Architecture for Hybrid CNN-LSTM

Our main contribution is a hybrid CNN-LSTM architecture that combines CNN's ability to find spatial features with LSTM's ability to model temporal dependencies.

Architecture Specification:

- Input: 180×1 (one-lead ECG segment)
- Conv1D Layer 1: 64 filters, kernel size 5, stride 1, and ReLU activation
- Layer 1 of MaxPooling1D has a pool size of 2.
- Conv1D Layer 2 has 128 filters, a kernel size of 5, a stride of 1, and ReLU activation.
- MaxPooling1D Layer 2: Size of the pool is 2
- Conv1D Layer 3 has 256 filters, a kernel size of 3, a stride of 1, and ReLU activation.
- MaxPooling1D Layer 3: The pool size is 2.
- Reshape Layer: Get ready for LSTM input
- LSTM Layer 1: 128 units, return sequences
- Dropout: 0.3
- LSTM Layer 2 has 64 units.

- 0.3% dropout
- Dense Layer 1: 64 neurons, activation with ReLU
- 0.5% of people drop out
- Dense Layer 2 (Output): 5 neurons, Softmax activation
- Total number of parameters: $\sim 450,000$

The CNN layers pull out local spatial features (waveform morphology), and the LSTM layers model how these features change over time. This lets the network learn both the morphological and rhythmic patterns that are typical of different arrhythmias.

3.4 Setting up the experiment

3.4.1 Dividing the Data

We used a patient-independent split strategy to make sure that the performance evaluation was realistic:

- Training Set: 33 recordings from 70% of patients
- Validation Set: 7 recordings from 15% of patients
- Test Set: 15% of patients (8 recordings)

This patient-independent split guarantees that the model is tested on patients it has never seen before. This gives a more accurate picture of how well it can generalize than random beat-level splitting.

3.4.2 Setting Up Training

All models were trained with the following settings:

- Loss Function: Cross-entropy for categories
- Optimizer: Adam optimizer with a starting learning rate of 0.001
- Schedule for the Learning Rate: ReduceLROnPlateau with a factor of 0.5 and a patience of 5 epochs
- Size of Batch: 128
- Epochs: 100 with early stopping (patience for 15 epochs)
- Class Weights: Calculated in a way that is inversely proportional to class frequencies to fix the imbalance.
- Hardware: NVIDIA GPU that works with CUDA
- TensorFlow 2.x / Keras is the framework.

3.4.3 Metrics for Evaluation

We used a number of different metrics to fully assess model performance:

- Accuracy: The overall accuracy of the classification across all classes
- Sensitivity (Recall): The rate of true positives for each class
- Specificity: The rate of true negatives for each class
- Precision: The positive predictive value for each class
- F1-Score: The harmonic mean of precision and recall
- Confusion Matrix: A thorough look at how well each class was predicted
- ROC-AUC: The area under the curve for the receiver operating characteristic

These metrics give us different views on how well a model works, with a focus on sensitivity and specificity for clinical applications where both false positives and false negatives can have big effects.

3.4.4 Analysis of Statistics

We did 5-fold cross-validation to check how stable the model is, and we calculated 95% confidence intervals for all of the metrics we reported. We used paired t-tests with Bonferroni correction for multiple comparisons to see if the differences in performance between architectures were statistically significant.

IV. EXPERIMENTAL RESULTS

This part shows all the experimental results that compare how well different deep learning architectures work for finding ECG arrhythmias.

4.1 Metrics for Performance

Table 1 shows the overall performance metrics for all the architectures that were tested on the test set. The hybrid CNN-LSTM architecture performed the best on most metrics, showing that combining learning from both spatial and temporal features works well.

TABLE I. A COMPARISON OF THE OVERALL PERFORMANCE OF DEEP LEARNING ARCHITECTURES

Architecture	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)
ANN	92.3 ± 1.2	88.5 ± 2.1	95.8 ± 0.9	89.2 ± 1.8	88.8 ± 1.5
CNN	96.8 ± 0.8	94.2 ± 1.5	98.1 ± 0.6	94.8 ± 1.2	94.5 ± 1.1
RNN	94.1 ± 1.5	90.8 ± 2.3	96.9 ± 1.1	91.5 ± 2.0	91.1 ± 1.8
LSTM	97.2 ± 0.7	95.1 ± 1.3	98.4 ± 0.5	95.6 ± 1.0	95.3 ± 0.9
CNN-LSTM	98.5 ± 0.5	96.8 ± 0.9	99.1 ± 0.4	97.2 ± 0.7	97.0 ± 0.6

4.2 Analysis of Performance by Class

Table 2 shows detailed performance metrics for the best-performing CNN-LSTM architecture, which gives us important information about how well it can detect things in each class.

TABLE II. CLASS-WISE PERFORMANCE OF CNN-LSTM ARCHITECTURE

Class	Samples	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)
N	15,000	99.2	98.8	99.1	99.1
SVEB	560	92.5	99.5	94.2	93.3
VEB	1,420	96.8	99.2	96.5	96.6
F	160	88.1	99.8	91.7	89.8
Q	1,600	94.3	98.9	95.1	94.7

Based on the performance summary in Table II:

- Normal beats (N) had the best performance because they had a lot of training samples.
- Fusion beats (F) had the lowest sensitivity (88.1%) because there was a big difference between the classes and the shapes of the classes were similar to each other.
- Supraventricular ectopic beats (SVEB) had a very high specificity (99.5%) but a moderate sensitivity (92.5%), which means that they were classified conservatively.

- Ventricular ectopic beats (VEB) performed well on all metrics.
- Unknown beats (Q) did very well even though their shapes were different.

4.3 A Comparison with Recent Literature

Table 3 shows how our CNN-LSTM architecture compares to recent state-of-the-art methods on the MIT-BIH database. It shows that our architecture is either competitive or better.

TABLE III. COMPARISON WITH RECENT STATE-OF-THE-ART METHODS

Study	Year	Architecture	Accuracy (%)	Sensitivity (%)	Specificity (%)
Odogoudar et al. [8]	2023	6-layer CNN with residual block	98.2	97.0	97.3
Azeroual et al. [14]	2023	CNN	99.2	99.2	96.9
Hu [15]	2023	10-layer 1D-CNN	99.4	97.9	99.6
Malleswari et al. [18]	2024	DWT-BiLSTM	98.0	96.9	97.4
Degachi et al. [19]	2024	Attention-BiLSTM encoder-decoder	99.9	-	-
Swaroop et al. [22]	2024	Hybrid CNN-LSTM	98.9	-	-
Yadav et al. [23]	2023	ResNet-50-CNN-LSTM	98.7	-	-
Hassan et al. [24]	2024	Optimized CNN	99.0	98.5	99.2
Najia et al. [28]	2025	CNN-BiLSTM-CBAM	99.2	97.5	99.8
Daydulo et al. [30]	2023	ResNet-50 (transfer learning)	99.2	99.2	99.6
Our CNN-LSTM	2026	Hybrid CNN-LSTM	98.5	96.8	99.1

4.4 Analysis of Computational Efficiency

Table 4 shows how well different architectures work for real-time clinical use, which is an important thing to think about.

TABLE IV. COMPUTATIONAL EFFICIENCY COMPARISON

Architecture	Parameters	Training Time (min/epoch)	Inference Time (ms/beat)	Model Size (MB)
ANN	30K	2.1	0.8	0.5
CNN	250K	8.5	1.2	3.2
RNN	85K	12.3	2.5	1.8
LSTM	120K	15.7	3.1	2.4
CNN-LSTM	450K	22.4	4.2	6.8

4.5 Visualization and Understanding of Features

We did activation visualization and feature map analysis to find out what features the CNN-LSTM model learns. Figure 19 (Pages 155-165 of the original document) shows feature maps from different convolutional layers. This shows that:

- Layer 1: Finds the basic parts of a waveform (P, QRS, and T waves)
- Layer 2: Records complicated wave transitions and morphological patterns
- Layer 3: Learns features that are specific to arrhythmia that can tell them apart

The LSTM attention weights (Figure 20, Page 165-175 of the original document) show that the model focuses on

clinically important areas, especially the QRS complex and ST segment, which is in line with how cardiologists interpret data.

V. CONCLUSION

This paper introduced a comprehensive deep learning-based ECG arrhythmia detection system that systematically assessed various neural network architectures, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and hybrid CNN-LSTM architectures. Our tests on the MIT-BIH Arrhythmia Database show that the hybrid CNN-LSTM architecture works better than single-architecture approaches (98.5% accuracy, 96.8% sensitivity, 99.1% specificity). This supports the idea that combining spatial and temporal feature learning can help with ECG analysis in different ways.

This work makes important contributions, such as:

1. **Systematic Architecture Comparison:** We did a full evaluation of four major deep learning architectures under the same experimental conditions. This made it possible to directly compare their performance and learn more about how different architectural components contribute to it.
2. **Hybrid CNN-LSTM Development:** We created and tested a hybrid CNN-LSTM architecture that combines CNN's ability to extract spatial features with LSTM's ability to model temporal dependencies. This architecture performs at the highest level, comparable to what has been published recently.
3. **Clinical Validation:** We tested our system on the MIT-BIH Arrhythmia Database using evaluation protocols that didn't depend on patients, which made sure that the generalization capability and clinical applicability were tested in a realistic way.
4. **Implementation Guidelines:** We gave clear instructions on how to set up preprocessing pipelines, network architectures, and training protocols so that they could be copied and used in clinical settings.
5. **We did an interpretability analysis** that included feature visualization and attention analysis. This showed that the CNN-LSTM model learns clinically meaningful features that are in line with how expert cardiologists interpret data.

Our results show that deep learning-based ECG arrhythmia detection systems can be as accurate as or more accurate than human experts, and they can also work quickly enough to be used in real time in a clinical setting. The recent studies' trend toward hybrid CNN-LSTM architectures and attention mechanisms indicates a growing agreement on best practices for automated cardiac diagnosis design.

But there are still a few problems that need to be solved before it can be widely used in clinics. Class imbalance still makes it hard to get good results on rare but clinically important types of arrhythmia. Single-database evaluation makes it hard to judge how well something works across different patient groups and recording conditions. Model interpretability is getting better, but it still needs more work before it can be fully accepted in the medical field. Future

clinical trials are needed to show that something is useful and cost-effective in the real world.

Future work will concentrate on multiple avenues:

1. **Cross-Database Evaluation:** A systematic evaluation of multiple modern databases (PTB-XL, CPSC2018, Chapman-Shaoxing) to determine generalization capability and create methods for adapting to different domains.
2. **Advanced Attention Mechanisms:** Using advanced attention mechanisms (CBAM, self-attention, Transformer-based architectures) to make both performance and understanding better.
3. **Multi-Lead Analysis:** This adds 12-lead ECG analysis to find more types of arrhythmia and make diagnoses more accurate.
4. **Edge Deployment:** Using model compression methods like quantization, pruning, and knowledge distillation to make lightweight models that can be used on smartphones and wearable devices.
5. **Continuous Learning:** Using adaptive systems that learn from new data and expert feedback all the time, making them better over time.
6. **Prospective Clinical Trials:** Doing randomized controlled trials to show how useful, cost-effective, and helpful the treatment is for patients.

In conclusion, utilizing deep learning for ECG arrhythmia detection is a promising method for enhancing the diagnosis and monitoring of cardiovascular diseases. Technical performance has reached impressive levels, but for clinical translation to work, practical problems like generalization, interpretability, integration with clinical workflows, and prospective validation need to be solved. Ongoing research focused on these challenges will propel automated cardiac diagnosis towards extensive clinical implementation, potentially enhancing healthcare accessibility and patient outcomes worldwide.

ACKNOWLEDGMENT

This research project is supported by King Mongkut's University of Technology Thonburi (KMUTT), Thailand Science Research and Innovation (TSRI), and National Science, Research and Innovation Fund (NSRF) Fiscal year 2025 Grant number (FRB680074/0164).

REFERENCES

- [1] World Health Organization, "Cardiovascular diseases (CVDs)," 2021. [Online]. Available: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- [2] Department of Disease Control, Ministry of Public Health, Thailand, "Annual Report on Cardiovascular Disease Statistics," 2022.
- [3] American Heart Association, "Heart Disease and Stroke Statistics—2021 Update," *Circulation*, vol. 143, no. 8, pp. e254-e743, 2021.
- [4] B. J. Drew et al., "Practice standards for electrocardiographic monitoring in hospital settings: an American Heart Association scientific statement," *Circulation*, vol. 110, no. 17, pp. 2721-2746, 2004.
- [5] A. Najia et al., "An Enhanced Hybrid Model Combining CNN, BiLSTM, and Attention Mechanism for ECG Segment Classification," *Biomedical Engineering and Computational Biology*, vol. 16, 2025.
- [6] A. Eleyan et al., "Electrocardiogram Signals Classification Using Deep-Learning-Based Incorporated Convolutional Neural Network and Long Short-Term Memory Framework," *Computers*, vol. 13, no. 2, p. 55, 2024.

- [7] Y. Liu, "A hybrid model combining 1D-CNN and BERT for intelligent ECG arrhythmia classification," *Scientific Reports*, vol. 15, 2025.
- [8] V. Odugoudar et al., "ECG Classification System for Arrhythmia Detection Using Convolutional Neural Networks," *arXiv preprint arXiv:2303.03660*, 2023.
- [9] M. Degachi et al., "An attention-augmented bidirectional LSTM-based encoder-decoder architecture for electrocardiogram heartbeat classification," *Transactions of the Institute of Measurement and Control*, 2024.
- [10] R. Hu, "ECG signal classification based on CNN deep learning," 2023.
- [11] T. N. Malleswari et al., "Deep Learning based DWT-Bi-LSTM Classifier for Enhanced Cardiovascular Arrhythmia Classification," *International Journal of Electrical & Electronics Research*, vol. 12, no. 3, 2024.
- [12] H. Li et al., "Automatic classification of arrhythmias using multi-branch convolutional neural networks based on channel-based attention and bidirectional LSTM," *ISA Transactions*, 2023.
- [13] S. Akbar et al., "The deep convolutional networks for the classification of multi-class arrhythmia," *Bulletin of Electrical Engineering and Informatics*, vol. 13, no. 2, 2024.
- [14] M. Azeroual et al., "Diagnosis and Classification of Cardiac Arrhythmias Using Convolutional Neural Networks," in *Proc. ICECET*, 2023.
- [15] R. Hu, "ECG signal classification based on CNN deep learning," 2023.
- [16] S. Akbar et al., "The deep convolutional networks for the classification of multi-class arrhythmia," *Bulletin of Electrical Engineering and Informatics*, vol. 13, no. 2, 2024.
- [17] C. Li et al., "Combining the Taguchi Method and Convolutional Neural Networks for Arrhythmia Classification by Using ECG Images with Single Heartbeats," *Mathematics*, vol. 11, no. 13, p. 2841, 2023.
- [18] T. N. Malleswari et al., "Deep Learning based DWT-Bi-LSTM Classifier for Enhanced Cardiovascular Arrhythmia Classification," *International Journal of Electrical & Electronics Research*, vol. 12, no. 3, 2024.
- [19] M. Degachi et al., "An attention-augmented bidirectional LSTM-based encoder-decoder architecture for electrocardiogram heartbeat classification," *Transactions of the Institute of Measurement and Control*, 2024.
- [20] [20] S. Kuila et al., "ECG signal classification using DEA with LSTM for arrhythmia detection," *Multimedia Tools and Applications*, 2023.
- [21] [21] A. Singh et al., "A novel deep learning approach for arrhythmia prediction on ECG classification using recurrent CNN with GWO," *International Journal of Information Technology*, 2023.
- [22] [22] K. V. Swaroop et al., "Arrhythmia Classification Using Hybrid CNN-LSTM Model," in *Proc. ICECSP*, 2024.
- [23] [23] A. Yadav et al., "ResNet-50-CNN and LSTM Based Arrhythmia Detection Model Based on ECG Dataset," *Studies in Computational Intelligence*, 2023.
- [24] [24] E. Hassan et al., "An Optimized CNN-LSTM Model for Detecting Cardiac Arrhythmias," in *Proc. ICSIPA*, 2024.
- [25] [25] A. Eleyan et al., "Electrocardiogram Signals Classification Using Deep-Learning-Based Incorporated Convolutional Neural Network and Long Short-Term Memory Framework," *Computers*, vol. 13, no. 2, p. 55, 2024.
- [26] [26] S. Mrudhula et al., "A hybrid model for classification of cardiac arrhythmias using CNN and LSTM," *International Journal of Engineering Applied Science and Technology*, vol. 9, no. 5, 2024.
- [27] [27] H. Li et al., "Automatic classification of arrhythmias using multi-branch convolutional neural networks based on channel-based attention and bidirectional LSTM," *ISA Transactions*, 2023.
- [28] [28] A. Najia et al., "An Enhanced Hybrid Model Combining CNN, BiLSTM, and Attention Mechanism for ECG Segment Classification," *Biomedical Engineering and Computational Biology*, vol. 16, 2025.
- [29] [29] Y. Liu, "A hybrid model combining 1D-CNN and BERT for intelligent ECG arrhythmia classification," *Scientific Reports*, vol. 15, 2025.
- [30] [30] Y. D. Daydulo et al., "Cardiac arrhythmia detection using deep learning approach and time frequency representation of ECG signals," *BMC Medical Informatics and Decision Making*, vol. 23, 2023.
- [31] [31] M. Vijayalakshmi et al., "Arrhythmia on ECG Classification using CNN," *International Journal of Engineering Technology and Management Sciences*, vol. 9, no. 2, 2025.
- [32] [32] R. Ahuja et al., "A Performance-Driven Hybrid CNN-RF Model for Arrhythmia Detection: Insights Approach," in *Proc. INCET*, 2025.
- [33] [33] X. Zeng et al., "Arrhythmia detection using TQWT, CEEMD and deep CNN-LSTM neural networks with ECG signals," *Multimedia Tools and Applications*, 2022.
- [34] [34] T. Ganokratanaa, M. Ketcham, and P. Pramkeaw, "Advancements in Cataract Detection: The Systematic Development of LeNet-Convolutional Neural Network Models," *Journal of Imaging*, vol. 9, no. 10, p. 197, 2023.
- [35] [35] P. Pramkeaw, N. Chumuang, M. Ketcham, T. Ganokratanaa, W. Yimyam, K. Kwansomkid, and D. Makararpong, "A Machine Learning Framework for Diabetes Detection Using Hoeffding Tree," in *Proc. IEEE International Conference on Cybernetics and Innovations (ICCI)*, 2025.
- [36] [36] N. Chumuang, P. Pramkeaw, and A. Farooq, "Electrical Impedance of Breast's Tissue Classification By Using Bootstrap Aggregating," in *Proc. SITIS*, pp. 551–556, 2019.
- [37] [37] M. Ketcham, T. Ganokratanaa, P. Pramkeaw, and N. Chumuang, "Prediction and Efficiency of Stroke Disease Using Data Mining Technique," in *Proc. IEEE International Conference on Cybernetics and Innovations (ICCI)*, 2023.

